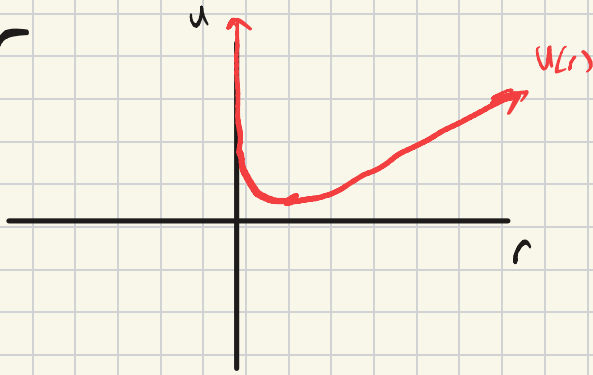


3a. $U(r) = U_0 \left(\frac{r}{R} + \lambda^2 \frac{R}{r} \right)$

as $r \rightarrow 0$; $U(r)$ will approach infinity exponentially as the $\lambda^2 \frac{R}{r}$ value takes over

as $r \rightarrow \infty$; $U(r)$ will increase linearly as the $\frac{r}{R}$ value takes over



3b. $\frac{dU}{dr} = U_0 \left(\frac{1}{R} - \lambda^2 \frac{R}{r^2} \right)$

$$0 = \frac{1}{R} - \lambda^2 \frac{R}{r_c^2}$$

$$\lambda^2 \frac{R}{r_c^2} = \frac{1}{R}$$

$$r_c^2 = \lambda^2 R^2$$

$$r_c = \lambda \cdot R$$

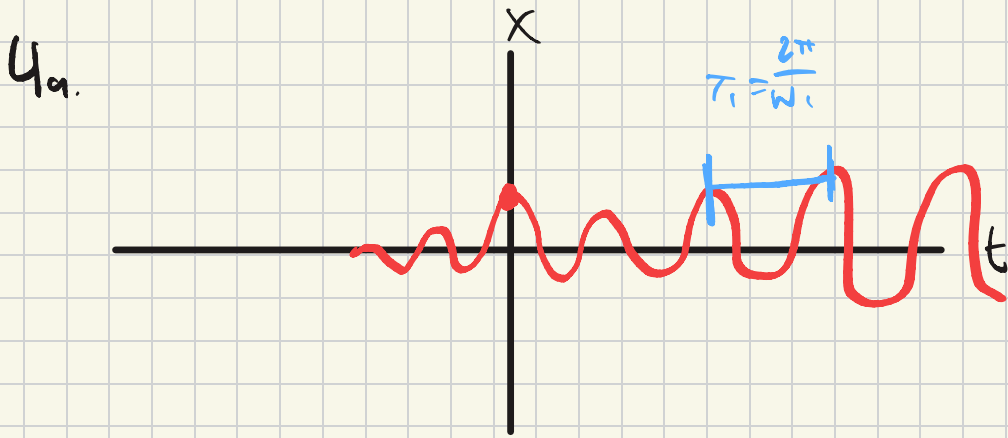
$$3c. U(x) = U(r_c) + U'(r_c)x + \frac{1}{2}U''(r_c)x^2$$

$$U''(r) = U_0(0 - \lambda^2 R(-2r^{-3})) = \frac{2U_0\lambda^2 R}{r^3}$$

$$\textcircled{a} r = r_c = \lambda R :$$

$$K = U''(r_c) = \frac{2U_0\lambda^2 R}{(\lambda R)^3} = \frac{2U_0}{\lambda R^2}$$

$$\omega = \sqrt{\frac{K}{m}} = \sqrt{\frac{2U_0}{m\lambda R^2}}$$



Time interval is between maximas

4b. Zero crossings @ $x(t) = 0$

$$Ae^{-\beta t} \neq 0 \text{ so } \cos(\omega_1 t - \delta) = 0$$

$$\cos = 0 \text{ @ } \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$\omega_1 (t_b - t_a) = \frac{3\pi}{2} - \frac{\pi}{2} = \pi$$

$$\Delta t = \pi / \omega_1$$

$$T_1 = \frac{2\pi}{\omega} \text{ so } T_1 = 2 \cdot \Delta t$$

$$4c. \beta = \frac{1}{2} \omega_0 \quad \omega_1 = \sqrt{\omega_0^2 - \beta^2}$$

$$\omega_1 = \sqrt{\omega_0^2 - \left(\frac{1}{2}\omega_0\right)^2} = \sqrt{\frac{3}{4}\omega_0^2} = \frac{\sqrt{3}}{2}\omega_0$$

$$\tau_1 = \frac{2\pi}{\omega_1} = \frac{2\pi}{\frac{\sqrt{3}}{2}\omega_0} = \frac{4\pi}{\sqrt{3}\omega_0}$$

$$t + \tau_1 / t \rightarrow \frac{A e^{-\beta(t+\tau_1)}}{A e^{-\beta t}} = e^{-\beta \tau_1}$$

$$\text{ratio: } e^{\left(-\frac{\omega_0}{2} \cdot \frac{4\pi}{\sqrt{3}\omega_0}\right)} = e^{\left(-\frac{2\pi}{\sqrt{3}}\right)}$$

$$e^{\left(-\frac{2\pi}{\sqrt{3}}\right)} = 0.0266 \Rightarrow 2.66\% \text{ decay}$$